**Jose Tupayachi, Ph.D. Senior Developer** | `jtupayachi.github.io`
Industrial and Systems Engineering | University of Tennessee, Knoxville | Oak Ridge National Laboratory

## EDUCATION

Ph.D. Industrial Engineering, University of Tennessee, Knoxville, May 2024–Present
M.S. Industrial Engineering, University of Tennessee, Knoxville, May 2024
B.S. Industrial Engineering, Pontifical Catholic University of Peru, 2020

## PROFESSIONAL POSITIONS

**Research Fellow**, Oak Ridge National Laboratory / ORISE, Mar 2025–Present. Developing advanced computer vision, multimodal AI, and scientific computing systems for materials science applications; building Vision Transformer (ViT) architectures and AI-assisted characterization pipelines for SEM imagery; designing explainable ML models for extreme weather analysis and geospatial AI interfaces; IP disclosures filed.

**Graduate Research Assistant**, University of Tennessee, Knoxville, Jul 2024–Present. Leading research on AI-powered freight transportation digital twins (RECOIL, DOE ARPA-E), mobile health applications (SmartShots, ACTAPP), and cancer staging automation; advising junior researchers.

Teaching Assistant – IE406/408: Simulation, UT, Knoxville, Spring 2022. Supported instruction in discrete-event simulation; led lab sessions, assisted with AnyLogic modeling assignments, and graded coursework.

Teaching Assistant – IE483: Reliability Engineering, UT, Knoxville, Fall 2022. Assisted with reliability analysis topics including Weibull distributions, failure mode analysis, and fault tree construction; held office hours and graded exams.

**Data Engineer**, Indra, Mar–Aug 2022. Designed and maintained ETL/ELT pipelines supporting batch and streaming workloads in a client-service delivery setting; built automated data quality checks (freshness, completeness, validity) and Git-based CI/CD workflows feeding warehouse and lakehouse layers for government and enterprise clients.

**Business Intelligence Analyst**, Globokas, Jan–Mar 2022. Developed BI dashboards, semantic models, and KPI reporting systems; integrated data from external systems and APIs and translated business problems into data-driven analyses for decision-making across units.

**Data Analyst Trainee**, Enel Distribución, Nov 2020–Dec 2021. Analyzed energy distribution data to identify consumption patterns and operational inefficiencies; developed predictive models and automated reporting tools.

**Developer – Portfolio Valuation**, GPS Management, Aug–Nov 2020. Implemented machine learning models for non-performing loan recovery and portfolio valuation; automated scoring workflows reducing manual processing time.

**Professional IT Intern**, Superintendencia de Banca, Seguros y AFP, Sep–Oct 2020. Supported IT infrastructure and regulatory data management; assisted in database maintenance and internal reporting for financial supervision operations.

**IT Intern**, Administradora Clínica Ricardo Palma, Feb–Aug 2019. Developed and maintained internal healthcare information systems; supported electronic health record integration and database administration for clinical operations.

## EXPERIENCE

Jose Tupayachi deploys advanced computer vision, multimodal AI, and scientific computing systems for materials science and engineering applications, developing Vision Transformer (ViT) architectures, foundation vision models, and AI-assisted characterization pipelines for scanning electron microscopy (SEM) imagery—enabling automated materials analysis, microstructure characterization, defect detection, feature extraction, and accelerated materials discovery workflows. By integrating large language models, multimodal foundation models, retrieval-augmented generation (RAG), and domain-specific scientific knowledge, he builds intelligent research platforms that transform complex experimental and simulation data into actionable insights for scientists and engineers. His work spans explainable machine learning for extreme weather prediction, geospatial AI interfaces, AI-powered freight transportation digital twins, and mobile health applications. He has published in leading journals and conferences including *Computers and Electrical Engineering*, *Health Informatics Journal*, ANNSIM, and ACM SIGSPATIAL, and has filed intellectual property disclosures at Oak Ridge National Laboratory.

## RELATED FUNDING SUPPORT

**Oak Ridge National Laboratory** Mar 2025–Present
*Role:* Research Fellow. Deployed advanced computer vision, multimodal AI, and scientific computing systems for materials science and engineering applications. Developed Vision Transformer (ViT) architectures, foundation vision models, and AI-assisted characterization pipelines for scanning electron microscopy (SEM) imagery, enabling automated materials analysis, microstructure characterization, defect detection, feature extraction, and accelerated materials discovery workflows. By integrating large language models, multimodal foundation models, retrieval-augmented generation (RAG), and domain-specific scientific knowledge, built intelligent research platforms that transform complex experimental and simulation data into actionable insights for scientists and engineers.
*Funding:* DOE Office of Science (SC), Fusion Energy Sciences (FES) – Fusion Materials Sciences Program; DOE Office of Environmental Management (EM), Nuclear Safety Research and Development (NSRD).

- Developed AI-agent workflows for materials discovery, scientific knowledge extraction, and literature exploration.
- Built computer vision, Vision Transformer (ViT), and foundation vision models for scanning electron microscopy (SEM) image characterization and materials science applications.
- Applied multimodal AI systems to automate microstructure analysis, defect identification, phase characterization, and feature extraction from scientific imagery.
- Developed AI-driven characterization pipelines supporting fusion materials research and advanced materials performance evaluation.
- Designed explainable machine learning models for extreme weather event prediction using atmospheric, environmental, and sensor datasets.
- Developed geospatial AI interfaces enabling natural-language interaction with spatial datasets and advanced analytics workflows.
- Created automated dashboard generation and scientific decision-support systems using large language models, retrieval-augmented generation (RAG), and modern web technologies.

**Intellectual Property Disclosures** | Oak Ridge National Laboratory 2025–Present
*Automated Geospatial Analysis Interface* – engineered an end-to-end interface driven by large language models, enabling non-expert users to query, process, and visualize spatial datasets through natural language. *IP disclosure filed, ORNL, 2025.*

**RECOIL** – Cognitive Freight Transportation Digital Twin for Resiliency and Emission Control Through Optimizing Intermodal Logistics Jul 2024–Present Designed ontology-guided optimization models for large-scale freight transportation networks, integrating data from GIS-based transport modes (road, rail, waterways) to minimize costs and emissions.
Leveraged Convolutional Neural Networks and Graph Neural Networks to analyze the impacts of weather, traffic, and demand on EV charging behavior, enabling data-driven, real-time optimization.
Applied large language models to power a chatbot providing domain-specific trade-off analysis between cost, time, and $CO_2$ emissions for non-technical users.
Designed a real-time feedback loop for digital twins using simulated sensor data and scraping techniques to continuously optimize intermodal system performance.
Investigated BERT and sentence transformers for domain literature QA with accurate and reproducible responses.
Collaborated with Oak Ridge National Laboratory researchers to create industry-ready solutions.
Developed the unified UI using Flutter at `https://m1.recoil.ise.utk.edu`.
*Funding Agency:* U.S. Department of Energy ARPA-E *Project:* #DE-AR0001780

**SmartShots** – Cross-Platform Application to Improve Childhood Vaccination Rates in Tennessee Dec 2023–Dec 2024 Enhanced vaccination tracking with real-time data updates, notifications, and alerts; developed a scalable Laravel backend and cross-platform Dart/Flutter frontend.
Integrated community health information for real-time access to nearby vaccination providers; collaborated with the Tennessee Department of Health.
`https://play.google.com/store/apps/details?id=com.ilab.smartshots`
`https://apps.apple.com/us/app/smartshots-tn/id6526502640`
*Funding Agency:* Tennessee Department of Health

**Active Caregiver's Toolkit (ACTAPP)** – Mobile App for Physical Activity Among Rural Appalachian Caregivers at Risk for CVD Jul 2024–Present Developed the ACT APP targeting cardiovascular disease risk reduction through physical activity interventions; implemented Dart 3.0, Material Design, and state management using GetX.
Integrated ChatBot for FAQ answering and page/interaction tracking metrics for data analysis.
`https://testflight.apple.com/join/PehxA8aW`
`https://play.google.com/store/apps/details?id=com.ilabutk.fe_rich`
*Funding Agency:* Hillman Emergent Innovation (HEI)

## HONORS AND AWARDS

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| 2025–2026 | **IISE Future Faculty Fellows**, USA – Selected |
| 2024 | **Best Paper Award**, CIE51 (51st Conference on Computers and Industrial Engineering), Sydney, Australia<br>*"Emerging AI and Cognitive Digital Twin Technologies Towards Low-Carbon Multimodal Freight Transport Systems"* |
| 2024 | **IISE DAIS Student Mobile App Competition – Winner**, SmartShots Project, Montreal<br>`https://www.iise.org/Details.aspx?id=33697` |
| 2024 | **HIDA Helmholtz Visiting Researcher**, Karlsruhe Institute of Technology (KIT), Germany – Awarded not taken |
| 2022–2024 | **Holiday Fellowship**, University of Tennessee, Knoxville (2022, 2023, 2024) |

## SELECTED PUBLICATIONS

TH Wyatt, S Lowe, J Tupayachi, X Li, CA McNeely, X Wang, PD Taylor et al., **Design and usability testing of SmartSHOTS: A mobile app to reduce vaccine barriers for children 0–24 months**, *Health Informatics Journal 32(1)* (2026).
J Tupayachi, AN Khan, X Li, **Scalable decentralized prognostics for industrial systems under data heterogeneity**, *Computers and Electrical Engineering 133, 111023* (2026).
J Tupayachi, H Xu, OA Omitaomu, MC Camur, A Sharmin, X Li, **Towards next-generation urban decision support systems through AI-powered construction of scientific ontology using large language models**, *Smart Cities 7(5), 2392–2421* (2024).
J Tupayachi, MM Ferguson, X Li, **A Simulation-Based Real-Time Deep Reinforcement Learning Approach for Fighting Wildfires**, *ANNSIM 2024, 1–12* (2024).
H Xu, X Li, J Tupayachi, JJ Lian, OA Omitaomu, **Automating Bibliometric Analysis with Sentence Transformers and RAG**, *Proc. 2nd ACM SIGSPATIAL Workshop on Advances in Urban-AI* (2024).
X Li, J Tupayachi, A Sharmin, M Martinez Ferguson, **Drone-aided delivery methods, challenges, and the future: A methodological review**, *Drones 7(3), 191* (2023).
J Tupayachi, L Silva, **Better Efficiency on Non-performing Loans Debt Recovery and Portfolio Valuation Using Machine Learning Techniques**, *POMS Lima, Peru* (2022).

**Pre-Prints**
X Li, H Xu, J Tupayachi, O Omitaomu, X Wang, **Empowering Cognitive Digital Twins with Generative Foundation Models: Developing a Low-Carbon Integrated Freight Transportation System**, *arXiv:2410.18089* (2024).
J Tupayachi, MC Camur, K Heaslip, X Li, **Predicting Electric Vehicle Charging Station Demand: Spatial-Temporal GNNs with Integrated Weather and Traffic Data**, *arXiv:2510.09048* (2024).
J Tupayachi, X-Y Yu, X Li, Z Liu, K Birdwell, **Explainable Machine Learning for Extreme Weather Event Prediction at the Oak Ridge Reservation**, *SSRN 6409258* (2025).

**Submitted Papers**
H Xu, Y Sun, J Tupayachi, O Omitaomu, S Zlatanova, X Li, **Towards Autonomous Urban Logistics Optimization via Generative AI and Agentic Digital Twins**, *Computers & Industrial Engineering* (ID: CAIE-D-25-05889).
J Tupayachi et al., **CliniSense AI for Automated Clinical Skills Assessment with Real-Time Feedback**, *npj Digital Medicine*.
H Xu, J Tupayachi, X-Y Yu, **Context-Aware Visual Prompting: Automating Geospatial Web Dashboards with LLMs and AI Agents**, *Int. J. Applied Earth Observation and Geoinformation* (ID: JAG-D-25-04850).
J Tupayachi, MC Camur, K Heaslip, X Li, **Spatio-Temporal GCNs for EV Charging Demand Forecasting**, *Applied Energy* (ID: 30181).

**Upcoming Publications**
Automated Control: The Tragedy of the Commons in End-to-End AI-Driven Systems
Literature Survey for New Materials Discovery Powered by AI Agents
Vision Transformer-Based Quality Assurance Engine for Wind Speed and Direction Classification
Automated End-to-End Geospatial Analysis: An AI-Driven Interface

## TECHNICAL SKILLS

**Languages:** Python, SQL, R, SAS, STATA, SPSS, Bash, Dart, PHP, Java, C++, CUDA. **Data Engineering:** Delta Lake, Databricks Workflows, Unity Catalog, Spark, dbt, Airflow, Prefect; batch and streaming ETL/ELT, data quality, and monitoring/alerting. **Data Platforms:** Snowflake, lakehouse and data-warehouse architecture, query authoring, and performance optimization. **Analytics:** Statistical modeling, clustering, cross-sectional and panel-data regression; analytics and ML data pipelines. **Frameworks:** Django, Flask, TensorFlow, PyTorch, Flutter, Laravel. **IoT:** Arduino, ESP32. **Databases:** PostgreSQL, MongoDB, MySQL, SQL Server. **DevOps:** Docker, Git, Jenkins, CI/CD, AWS, KVM, Slurm. **Optimization:** Gurobi, CPLEX, AnyLogic, NetworkX; model tuning and calibration. **Integration:** REST and GraphQL APIs; data-privacy-aware design. **GIS:** WMS, GeoServer, OverPass API.

## SYNERGISTIC ACTIVITIES

**Community Service. University of Tennessee Graduate Student Senate** – Senator, Industrial and Systems Engineering, 2025. Represented students, advocated for academic and professional development opportunities, and fostered graduate student engagement.

**Journal & Conference Reviewer.** PLOS ONE; Information Systems Frontiers (Springer); IUI 2026 (IEEE/ACM); Journal of Asian Architecture and Building Engineering (Elsevier); IEEE Internet of Things Magazine; Software: Practice and Experience (Wiley); IISE Transactions (Taylor & Francis); Optimization Letters (Springer Nature); Journal of Robotics (Wiley); Journal of Sensors (Wiley); Winter Simulation Conference 2025; IISE Annual Conference 2025; IEOM 2025.

**Conference Presenter.** Federated Learning Fault Detection: Towards a Decentralized ML Framework – IISE 2023. Rule-based Automated Cancer Staging from Scanned Pathology Reports – IISE 2024. Empowering Simulation Modeling: An Automated Ontology Framework Enhanced by LLMs – INFORMS 2024. Conversational Geographic Question Answering: LLMs & Continuous RAG – SIGSPATIAL 2024.